

Docker + PostgreSQL Command Reference

This document lists all Docker commands used while setting up and managing PostgreSQL for local development with FastAPI. Each command is explained with its purpose and when it is typically used.

docker compose up -d

Starts PostgreSQL using docker-compose in detached mode. Creates container, network, and volume if they do not exist. Used for initial startup and regular use.

docker compose down

Stops and removes the PostgreSQL container and network but keeps volumes. Used when restarting services without losing data.

docker compose down -v

Stops containers and deletes volumes. This completely wipes database data. Used when resetting credentials or database state during development.

docker ps

Lists all running Docker containers. Used to confirm PostgreSQL is running and to check container names.

docker exec -it auth-db psql -U myuser -d authdb

Opens an interactive PostgreSQL shell inside the running container. Used to inspect tables and query database contents directly.

docker exec -it auth-db psql -U myuser -l

Lists all databases present inside the PostgreSQL server. Used to verify database creation and debug missing database errors.

docker exec -it auth-db printenv | findstr POSTGRES

Displays PostgreSQL-related environment variables inside the container. Used to debug username, password, and database mismatches.

docker logs auth-db

Shows PostgreSQL container logs including startup messages and errors. Used when PostgreSQL fails to start or behaves unexpectedly.

docker stop auth-db

Gracefully stops the PostgreSQL container without deleting it. Used to temporarily shut down the database.

docker start auth-db

Starts a previously stopped PostgreSQL container while preserving data. Used to resume database operations.